

Insurance Cost Prediction Project Summary

This project focuses on predicting medical insurance charges using Machine Learning algorithms. The model predicts insurance cost based on factors such as age, BMI, smoking habits, number of children, gender, and region.

Project Aim

The aim of this project is to build a Machine Learning model that can accurately predict medical insurance charges using customer demographic and health-related information.

Dataset Features

- Age
- Gender
- BMI (Body Mass Index)
- Number of Children
- Smoker Status
- Region
- Insurance Charges (Target Variable)

Project Workflow

1. Data Collection and Loading
2. Data Cleaning and Duplicate Removal
3. Exploratory Data Analysis (EDA)
4. Feature Encoding using One-Hot Encoding
5. Train-Test Split
6. Model Training
7. Model Evaluation using MAE, MSE, RMSE, and R² Score
8. Model Comparison

Machine Learning Models Used

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

Model Performance Comparison

Model	R ² Score
Linear Regression	0.8069
Decision Tree Regressor	0.8189
Random Forest Regressor	0.8797

Conclusion

The project successfully predicted medical insurance charges using Machine Learning techniques. Among all models, the Random Forest Regressor performed best with the highest R² Score of 0.8797, indicating strong prediction

accuracy. The project demonstrates how Machine Learning can help insurance companies estimate customer insurance costs more efficiently.

Future Improvements

- Use larger real-world datasets
- Apply advanced boosting algorithms like XGBoost
- Deploy the model using Flask or Streamlit
- Integrate the model into web applications